Total Pages: 3

Scheme of Valuation/Answer Key

(Scheme of evaluation (marks in brackets) and answers of problems/key)

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIFTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER **2021**

Course Code: MCN 301

Course Name: DISASTER MANAGEMENT

Max. Marks: 100 Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

		Marks
1	Greenhouse effect relevance - 1.5 . Adverse effects -1.5	3
2	South West and North East(1.5 marks each)	3
3	Names of Three spatial characteristics, Two temporal characteristics & magnitude -(1.5.marks) sources – base maps, remotely sensed images, field data & the concepts-(1,5marks)	3
4	Principle for qualitative risk assessment (1.5marks) Risk matrix/Riak assessment matrix concept (1.5 marks)	3
5	6 types of disaster response (0.5mark each)	3
6	Six international relief organizations	3
7	Stating the difference through defimitions and explanations	3
8	Structural measures in capacity building (1.5 marks) Nonstructural measures in capacity building (1.5 marks)	3
9	Explanation of Legislations in India on disaster management	3
10	Explanation - Interrelation of National Disaster Management Policy with other national policies	3

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

11	a)	Diagram with names and distances marked = (1 mark) features of layers of atmosphere-7 marks	8
	b)	a) Disaster (2 marks) b) Hazard (2 marks) c) Risk (2 marks)	6

12	a)	Crisis counselling (4 marks). Necessity of counselling (4 marks)	8
	b)	Reasons : man made -CFCs/BFCs (halocarbon refrigerants, solvents, propellants, and foam- blowing agents), N_2O/NO (soil bacteria, use of fertilisers, jet aircrafts) , production of GH gases through various possibilities (4marks) Individual Prevention - Instead of using chemicals shift to natural methods/ eco-friendly products to get rid of pests and for cleaning acticvities. The vehicles emit a large amount of greenhouse gases that lead to global warming as well as ozone depletion. Therefore, the use of vehicles should be minimized as much as possible or use EV. Maintain air conditioners, as their malfunctions cause CFC to escape into the atmosphere. (Any 2 such suggestions – 2 marks)	6

Module -2

13	a)	Concept of hazard mapping -2GIS and participatory hazard mapping(3 marks each) GIS and participatory hazard mapping(3 marks each)	8
	b)	Societal risk = av. no. of deaths per year = 4407 (3 marks) Individual risk = yearly deaths per person = 4407/3.33cr (3 marks)	6
14	a)	Four different types of vulnerability(1.5 mark each), Four socio economic indicators -2 marks	8
	b)	Empirical & Analytical – (3 marks each)	6

Module -3

15	a)	Four core elements of disaster risk management(2.5 marks each)	10
	b)	Two requirements for effective disaster response(2 marks each)	4
16	a)	Disaster risk reduction definition (2 marks). Three measures for disaster risk reduction(2 marks each).	8
	b)	Relief(2 marks) Principles guiding relief(4 marks)	6

Module -4

17	a)	Four effective ways of promoting stakeholder participation in disaster risk reduction(4 marks)	8

		Benefits(4 marks)	
	b)	Three basic steps in participatory stakeholder engagement(2 marks each)	6
18	a)	Capacity building definition – 1 mark . Explanation -2 marks	10
		Importance of capacity assessment(4 marks)	
		Explain the methods of assessing capacity(3 marks)	
	b)	Barriers to effective communication - Any 4	4

Module -5

19	a)	Five disaster types(2 marks each)	10
	b)	Objectives of national disaster management policy (2 marks)	4
		main elements of national disaster management policy(2 marks)	
20	a)	Targets of Sendai Framework for disaster risk reduction (2 marks)	8
		Priorities (2 marks)	
		Guiding principles (4 marks)	
	b)	national, state, district -2 marks each	6

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